Abdullah Chabaytah

514-705-5932 | a.chabaytah@mail.utoronto.ca | linkedin.com/in/achabaytah | github.com/maniic | maniic.github.io

Education

University of Toronto, St. George — Toronto, ON
HBSc Computer Science, Math & Statistics — *Expected Grad: 05/2028*

- **Relevant Coursework:** *Data Structures & Algorithms, Theoretical Computation, Software Design, Systems Programming, Multivariable Calculus, Calculus with Proofs, Probability, Statistics & Data Analysis*

Experience

Google Developer Groups — 06/2026 – Present
Technology Executive — *University of Toronto, St. George*

- Designed and shipped a responsive website for GDG @ UofT St. George using **React**, **TypeScript**, and **HTML/CSS**, serving **1000+** students
- Built reusable **frontend** components to showcase events, resources, and initiatives, cutting page load times by **30%** through code splitting and asset optimization
- Integrated an LLM-powered chatbot into the GDG website using the **Gemini** API, answering common member questions about events, workshops, and resources

Sports Up — 05/2025 – 09/2025
Software Developer Intern — *Toronto, ON*

- Engineered a cross-platform mobile application using **React Native** & **Firebase**, achieving **99% uptime** across **100+ users** under sustained loads of **50 requests/sec** via real-time data synchronization & optimized query batching
- **Reduced crash rates by 35%** & **increased weekly active users by 20%** by refactoring backend architecture and rebuilding **UI/UX** components to eliminate state management bottlenecks
- Implemented end-to-end testing frameworks using automated test suites to enforce deployment quality gates, cutting regression failures across **3** major feature releases

Outlier AI — 01/2025 – Present
Data Labeling Contractor — *Remote*

- Annotated & validated **NLP** and **computer vision** training datasets, applying domain-specific labeling schemas to improve model precision across production **ML** pipelines

Projects

Predictive Maintenance | *Python, PyTorch, MLflow, Scikit-learn, Pandas*

- Trained deep learning architectures including **Bidirectional LSTM, Temporal CNN, and Transformer** using **PyTorch Lightning** to predict jet engine failure on **NASA** benchmark data, achieving **11.71 RMSE**
- Built end-to-end data pipelines processing sensor channels across multiple fault-mode datasets, implementing normalization, sequence windowing, and automated data quality validation
- Instrumented **MLflow** to log hyperparameters, metrics, and model artifacts across **100+** training runs, enabling reproducible experiment tracking and model comparison

Sentiment Forecasting | *Python, PyTorch, Hugging Face, Scikit-learn, Pandas*

- Built a quantitative forecasting pipeline achieving a **2.34 Sharpe ratio** and **-2.5% max drawdown** over **180-day** backtests by fusing **FinBERT** outputs with topological & momentum features
- Trained and evaluated **Logistic Regression, XGBoost, and Ensemble** classifiers using time-series aware cross-validation to eliminate look-ahead bias, selecting the optimal model via **Sharpe ratio** on held-out data
- Implemented **Topological Data Analysis** using 4-dimensional delay embeddings on 60-day rolling windows to extract market regime signals absent from traditional indicators

Technical Skills

Languages: Python, TypeScript/JavaScript, C/C++, Java, Golang, R, Kotlin, Assembly, Rust, SQL, HTML/CSS
Frameworks & Libraries: React, Node.js, TensorFlow, PyTorch, FastAPI, Flask, Django, Pandas, NumPy, Matplotlib
Systems & Tools: Docker, PostgreSQL, Redis, Git, Unix/Linux, CI/CD, VS Code, PyCharm, IntelliJ